

# Mathematical Analysis II Notes

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## 8 空间解析几何

### Linear Algebra and Analytic Geometry

#### 8.0.1 Inner Product

Let  $V$  be a real vector space, an inner product on  $V$  is a map.

$$\begin{aligned} g : V \times V &\rightarrow \mathbb{R} \\ (u, w) &\mapsto g(u, w) \end{aligned}$$

satisfying

1. **Bilinearity:**

$$\begin{aligned} g(\lambda_1 v_1 + \lambda_2 v_2, w) &= \lambda_1 g(v_1, w) + \lambda_2 g(v_2, w) \\ g(v, \lambda_1 w_1 + \lambda_2 w_2) &= \lambda_1 g(v, w_1) + \lambda_2 g(v, w_2) \end{aligned}$$

2. **Symmetric:**  $g(v, w) = g(w, v), \forall (v, w) \in V$

3. **Positive Definite 正定:**  $g(v, v) \geq 0, \forall v, “=” \text{ iff } v = 0$

An inner product is also denoted by  $\langle \cdot, \cdot \rangle_g$ , or *angle*  $\langle \cdot, \cdot \rangle$ , or “ $\cdot$ ” if there is no confusion.

**Eg.**  $V = C[0, 1], \langle f, g \rangle \triangleq \int_0^1 f(x)g(x) dx$

For  $\dim_{\mathbb{R}} V = n < +\infty$ , we have the following classification results:

**Thm** (Classification of **Symmetric Bilinear Form**). *If  $\dim_{\mathbb{R}} V = n$ , then for any symmetric bilinear form  $g$  on  $V$ , there exists a basis  $(v_1, \dots, v_n)$  s.t.*

$$(g(v_i, v_j))_{1 \leq i, j \leq n} = \begin{bmatrix} 1 & & & & & & & & \\ & \ddots & & & & & & & \\ & & 1 & & & & & & \\ & & & -1 & & & & & \\ & & & & \ddots & & & & \\ & & & & & -1 & & & \\ & & & & & & 0 & & \\ & & & & & & & \ddots & \\ & & & & & & & & 0 \end{bmatrix} (r_+ \uparrow 1, r_- \uparrow -1, n - r_+ - r_- \uparrow 0)$$

where  $r_+ = \max \{ \dim_{\mathbb{R}} W \mid g|_W > 0 \text{ for a subsequence } W \}$ ,  $r_-$  similarly defined.

is called the positive/negative inertia index respectively.

**Thm** (Classification of Inner Product). *If  $\dim_{\mathbb{R}} V = n$ , then for any inner product  $g$  on  $V$ ,*

1. there exists a basis  $(v_1, \dots, v_n)$  s.t.

$$g(v_i, v_j) = \begin{pmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{pmatrix}, \text{ i.e. } g(v_i, v_j) = \delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases} \text{ (**Kronecker Symbol**)}$$

i.e. orthonormal basis w.r.t.  $g$  exists.

2. any basis  $(v'_1, \dots, v'_n)$  is orthonormal w.r.t.  $g$  iff.  $(v'_1, \dots, v'_n) = (v_1, \dots, v_n)P$  for some  $P \in O(n) = \{A \in M_{n \times n}(\mathbb{R}) \mid AA^T = I\}$ , **called the orthonormal group 正交群**

In coordinates

Let  $V$  be a  $n$ -dim real vector space, an ordered set of  $n$  linearly independent vectors is called an ordered basis, usually denoted by  $(v_1, \dots, v_n)$ .

Fix a particular choice of ordered basis  $(v_1, \dots, v_n)$ , we have one-to-one correspondence.

1.  $V \rightarrow \mathbb{R}^n$  **isomorphism 同构**

$$v \rightarrow x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \text{ coordinate vector (w.r.t. the ordered basis } v_1, \dots, v_n)$$

$$\text{s.t. } v = \begin{pmatrix} v_1 & \cdots & v_n \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

2.

$L(v) = \{\text{linear map from } V \text{ to } V\} \rightarrow M_{n \times n}(\mathbb{R})$  **isomorphism**

$\mathcal{A} \mapsto A$  representing the linear map  $V$  under the basis  $(v_1, \dots, v_n)$

$$\mathcal{A} \cdot (\mathcal{B} + \mathcal{C}) = \mathcal{A} \cdot \mathcal{B} + \mathcal{A} \cdot \mathcal{C} \quad A(B + C) = AB + AC$$

3.

$\{\text{Bilinear form on } V\} \rightarrow M_{n \times n}(\mathbb{R})$  **isomorphism**

$$g \mapsto (g(v_i, v_j))_{1 \leq i, j \leq n}$$

4.

$\{\text{Symmetric Bilinear form on } V\} \rightarrow \text{Sym}_{n \times n}(\mathbb{R}) = \{A \in M_{n \times n}(\mathbb{R}) \mid A^T = A\}$  **isomorphism**

$$g \mapsto (g(v_i, v_j))_{1 \leq i, j \leq n}$$

5.  $\{\text{inner product on } V\} \rightarrow \text{Sym}_{n \times n}^+(\mathbb{R}) = \{A \in \text{Sym}_{n \times n}(\mathbb{R}) \mid A > 0\}$

(5)  $\subseteq$  (4) **convex cone**  $\rightarrow g_1, g_2 \Rightarrow \lambda_1 g_1 + \lambda_2 g_2, \lambda_1 \lambda_2 \geq 0$ , but  $\lambda_1^2 + \lambda_2^2 \neq 0, \dots$

**Thm.** For any  $A \in \text{Sym}_{n \times n}(\mathbb{R})$ , there exists  $P \in GL(n, \mathbb{R}) = \{B \in M_{n \times n}(\mathbb{R}) \mid \det B \neq 0\} \rightarrow$   
**General Linear Group  $GL(n)$**

$$s.t. PAP^T = \begin{pmatrix} 1 & & & & & & \\ & \ddots & & & & & \\ & & 1 & & & & \\ & & & -1 & & & \\ & & & & \ddots & & \\ & & & & & -1 & \\ & & & & & & 0 \\ & & & & & & & \ddots \\ & & & & & & & & 0 \end{pmatrix}$$

**Thm.** For any  $A \in \text{Sym}_{n \times n}^+(\mathbb{R})$ , there exists  $P \in GL(n, \mathbb{R})$ , s.t.  $PAP^T = \begin{pmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{pmatrix}$

$$(\Rightarrow A = P^{-1} (P^{-1})^T)$$

**Thm** (Spectral Theorem / Orthogonal Diagonalization). For any  $A \in \text{Sym}_{n \times n}(\mathbb{R})$ , there exists  $P \in O(n)$  and  $\lambda_1, \dots, \lambda_n \in \mathbb{R}$ , s.t.  $PAP^T = \text{diag}(\lambda_1, \dots, \lambda_n) = A$  ( $\Rightarrow$  **If  $A > 0$ ,  $A = P^T \Lambda P = P^T \Lambda^{\frac{1}{2}} P \cdot P^T \Lambda^{\frac{1}{2}} P = B^2$ , where  $B^T = B$ ,  $B > 0$** )

### 8.0.2 Geometric notions induced from inner product $g$ on $V$

1. Length of a vector:  $v \in V, |v|^g = \sqrt{\langle v, v \rangle_g}$
2. Angle: for  $v, w \in V \setminus \{0\}$ , the angle between  $v$  and  $w$  is defined as

$$\theta(v, w) = \cos^{-1} \frac{\langle v, w \rangle_g}{|v|_g |w|_g}$$

(where  $\cos^{-1}$  is the inverse of  $\cos : [0, \pi] \rightarrow [-1, 1]$ )

(Cauchy-Schwarz Inequality:  $|\langle v, w \rangle_g| \leq |v|_g |w|_g$ )

If  $\theta(v, w) = \frac{\pi}{2}$ , we say  $v$  is perpendicular to  $w$  w.r.t.  $g$ , denoted by  $v \perp_g w$ , or  $v \perp w$  if no confusion.

3. Orthogonal projection: for  $\vec{b} \in V \setminus \{0\}$



$$\text{pr}_{\vec{b}} : V \rightarrow \langle \vec{b} \rangle$$

$$\vec{a} \mapsto \frac{\langle \vec{a}, \vec{b} \rangle_g}{|\vec{b}|_g} \frac{\vec{b}}{|\vec{b}|_g}$$

$$\text{pr}_{\langle \vec{b} \rangle^\perp} : V \rightarrow \langle \vec{b} \rangle^{\perp_g} \text{ orthogonal complement}$$

$$\vec{a} \mapsto \vec{a} - \text{pr}_{\vec{b}} \vec{a}$$

$$V = \langle \vec{b} \rangle \oplus \langle \vec{b} \rangle^{\perp_g}$$

4. Volume of “ $k$ -dimensional” parallelepiped 平行多面体 is defined inductively.

The set  $P_{v_1, \dots, v_k} = \left\{ \sum_{i=1}^k t_i v_i \mid 0 \leq t_1, \dots, t_k \leq 1 \right\}$  is called a parallelepiped spanned by  $(v_1, \dots, v_k)$  in  $V$ .

- $k = 1$ ,  $\text{Vol}_g(P_v) = |v|_g$
- $k = 2$ ,  $\text{Vol}_g(P_{v_1, v_2}) \triangleq \text{Vol}_g(P_{v_1}) \left| \text{pr}_{\langle v_1 \rangle^\perp} v_2 \right|_g$

$$\begin{aligned} \text{Vol}_g(P_{v_1, v_2})^2 &= |v_1|_g^2 \left( |v_2|_g^2 - \left( \frac{\langle v_1, v_2 \rangle_g}{|v_1|_g} \right)^2 \right) \\ &= |v_1|_g^2 |v_2|_g^2 - \langle v_1, v_2 \rangle_g^2 \\ &= \det \begin{pmatrix} \langle v_1, v_1 \rangle_g & \langle v_1, v_2 \rangle_g \\ \langle v_2, v_1 \rangle_g & \langle v_2, v_2 \rangle_g \end{pmatrix} \end{aligned}$$

$$v_2 = \text{pr}_{v_1} v_2 + \text{pr}_{v_1^\perp} v_2$$

$$|v_2|_g^2 = |\text{pr}_{v_1} v_2|_g^2 + |\text{pr}_{v_1^\perp} v_2|_g^2$$

$$\text{Thm. } \text{Vol}_g(P_{v_1, \dots, v_k}) = \sqrt{\det \left( \langle v_i, v_j \rangle_g \right)}$$

### 8.0.3 Orientation 定向

**Def** (Equivalence Relation). A subset  $S \subseteq X \times X = \{(x, y) \mid x \in X, y \in Y\}$  is called a binary relation on  $X$ . A binary relation  $S$  on  $X$  is called an **equivalence relation** 等价关系 on  $X$  if

1. Symmetry:  $(x, y) \in S \Rightarrow (y, x) \in S$
2. Transitivity:  $(x, y) \in S, (y, z) \in S \Rightarrow (x, z) \in S$

3. *Reflexivity*:  $(x, x) \in S, \forall x \in X$

It is usually denoted by “ $\sim_S$ ” or “ $\sim$ ”

**Def** (Equivalence Class). Let “ $\sim$ ” be equivalence relation on  $X$ , then for any  $x \in X$ , the set  $\{y \in X \mid y \sim x\} \subseteq X$  is called an equivalence class, it is denoted by  $[x]_{\sim}$ . Any  $y \in [x]_{\sim}$  is called a representative of  $[x]_{\sim}$ .  $X/\sim \triangleq \{[x]_{\sim} \mid x \in X\}$  is called the quotient set of  $X$  under the relation “ $\sim$ ”.

**Eg.** 1.  $\{(x, y) \in \mathbb{R}^2 \mid x > y\}$  (“ $>$ ”) is a binary relation on  $\mathbb{R}$ , but not an equivalence relation.

2.  $S = \Delta x = \{(x, x) \mid x \in X\}$  (“ $=$ ”) is an equivalence relation.

3. (Congruence) Fix  $n \in \mathbb{N}$

$$\{(x, y) \in \mathbb{Z} \times \mathbb{Z} \mid n \mid (x - y)\}$$

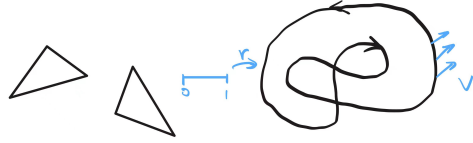
This equivalence relation is called “ $\text{mod } n$  congruence” ( $\equiv n$ )

$$\mathbb{Z}/\equiv n \left( \triangleq \mathbb{Z}_n \right) = \{[0]_{\equiv n}, \dots, [n-1]_{\equiv n}\}$$

$$\mathbb{Z}_2 = \{[0]_{\equiv 2}, [1]_{\equiv 2}\}$$

$[k]_{\equiv n} = \{l \in \mathbb{Z} \mid n \mid (l - k)\} = \{k + mn \mid m \in \mathbb{Z}\}$ , is called the “ $\text{mod } n$  congruence class of  $k$ ”.

**Eg.** 1.  $X = \{\text{triangle on } \mathbb{R}^2\} \cong \cong$



2. “Similarity” of matrix  $A \sim B \iff A = PBP^{-1}$  for some  $P \in GL(n, \mathbb{R})$

$$M_{n \times n}(\mathbb{R})/\sim = \{\text{Jordan canonical form}\}$$

**Def** (Orientation 定向, Oriented Vector Space 定向线性空间). Let  $\mathcal{B}_V = \{\text{Ordered basis of } V\}$  where  $V$  is a real vector space, define

$$(v'_1, \dots, v'_n) = (v_1, \dots, v_n)P$$

for some  $P \in GL(n, \mathbb{R})$ ,  $\det P > 0$  iff  $(v'_1, \dots, v'_n) \sim (v_1, \dots, v_n)$ .

This is an equivalence relation, and  $B_V/\sim = \{o_1, o_2\}$ , an element of  $B_V/\sim$  is called an orientation on  $V$ . Any vector space, together with a choice of orientation, is called an oriented vector space.

### 8.0.4 $\mathbb{R}^n$ : Euclidean Space

Equipped with the standard inner product

$$\langle x, y \rangle = \sum_{i=1}^n x_i y_i$$

where  $x = (x_1, \dots, x_n)^+, y = (y_1, \dots, y_n)^+ \in \mathbb{R}^n$ .

Under this inner product,  $(e_1, \dots, e_n)$ , where  $e_i = (0, \dots, 0, \overset{i}{1}, 0, \dots, 0)$ , is called the *standard orthonormal basis*.

**Rmk.**  $n = 2$ ,  $(e_1, e_2)$  is usually denoted by  $i, j$ ;

$n = 3$ ,  $(e_1, e_2, e_3)$  is denoted by  $i, j, k$

**Def** (Cross Product in  $\mathbb{R}^3$ ). Let  $x, y \in \mathbb{R}^3$ , define “ $\times$ ” via

$$\begin{aligned} x \times y &= \begin{vmatrix} i & j & k \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{vmatrix} = \begin{vmatrix} x_2 & x_3 \\ y_2 & y_3 \end{vmatrix} i - \begin{vmatrix} x_1 & x_3 \\ y_1 & y_3 \end{vmatrix} j + \begin{vmatrix} x_1 & x_2 \\ y_1 & y_2 \end{vmatrix} k \\ &= (x_2 y_3 - x_3 y_2) i + (x_3 y_1 - x_1 y_3) j + (x_1 y_2 - x_2 y_1) k \end{aligned}$$

call it the cross product of  $x$  and  $y$ .

**Prop.** 1.  $x \times y = (-y \times x)$  ( $x \times y = 0 \iff \{x, y\}$  linear dependent) (*Anti-Symmetric*)

2.  $(\lambda x + \mu y) \times z = \lambda x \times z + \mu y \times z$  (*Bilinear*)

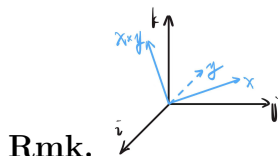
3.  $\langle x \times y, x \rangle = 0$   $\langle x \times y, y \rangle = 0$ ;  $(x, y, x \times y)$  determines the same orientation as  $(i, j, k)$

4.  $|x \times y| = \text{Area}(P_{x,y})$

**Pf.** 3.

$$\begin{aligned} (x, y, x \times y) &= (i, j, k) \begin{vmatrix} x_1 & y_1 & x_2 & y_2 \\ x_2 & y_2 & x_3 & y_3 \\ x_3 & y_3 & x_1 & y_1 \\ x_1 & y_1 & x_2 & y_2 \end{vmatrix} \\ &= |x \times y|^2 \end{aligned}$$

$\implies$  if  $\{x, y\}$  are linearly independent,  $(x, y, x \times y)$  has the same orientation as  $(i, j, k)$ .



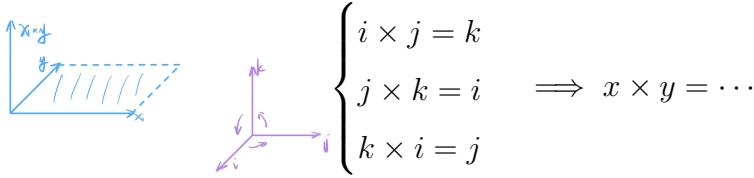
**Rmk.**

- The orientation denoted by  $(i, j, k)$  is called *Right-Hand orientation*;
- The orientation denoted by  $(i, j, -k)$  is called *Left-Hand orientation*;
- The ordered basis with the *Right-Hand orientation* is called a *Right-Hand system*.

4.

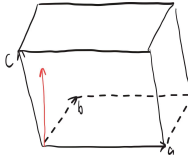
$$\begin{aligned}
 |x \times y|^2 &= (x_2y_3 - x_3y_2)^2 + (x_3y_1 - x_1y_3)^2 + (x_1y_2 - x_2y_1)^2 \\
 \text{Area}(P_{x,y}) &= \det \begin{pmatrix} \langle x, x \rangle & \langle x, y \rangle \\ \langle y, x \rangle & \langle y, y \rangle \end{pmatrix} \\
 &= (x_1^2 + x_2^2 + x_3^2)(y_1^2 + y_2^2 + y_3^2) - (x_1y_1 + x_2y_2 + x_3y_3)^2
 \end{aligned}$$

**Thm.** In  $\mathbb{R}^3$ , the cross product is uniquely determined by the four properties.



$$\begin{cases} i \times j = k \\ j \times k = i \\ k \times i = j \end{cases} \implies x \times y = \dots$$

**Def** (Mixed Product on  $\mathbb{R}^3$  混合积). Let  $a, b, c \in \mathbb{R}^3$

$$\langle a, b \times c \rangle = \langle b, a \times c \rangle = \langle c, a \times b \rangle = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$


is called the mixed product of  $a, b, c$ .

- $\langle c, a \times b \rangle$  measures the “**signed volume**” of  $P_{a,b,c}$ ;
- This equation also gives the formula for the volume of a parallelepiped in  $\mathbb{R}^3$ ;
- $\langle c, a \times b \rangle \begin{cases} > 0 & (a, b, c) \text{ is a Right-Hand system} \\ = 0 & \{a, b, c\} \text{ linearly dependent} \\ < 0 & (a, b, c) \text{ is a Left-Hand system} \end{cases}$

**Recall:**

$$\text{Vol}(P_{v_1, \dots, v_k}) = \sqrt{\det(\langle v_i, v_j \rangle)}, \quad v_1, \dots, v_k \in \mathbb{R}^n$$

If  $v_i = \sum_{s=1}^n e_s P_{s_i}$  ( $\iff (v_1, \dots, v_k) = (e_1, \dots, e_n) P$ ), then

$$\begin{aligned}
 \langle v_i, v_j \rangle &= \left\langle \sum_{s=1}^n e_s P_{s_i}, \sum_{t=1}^n e_t P_{t_j} \right\rangle = \sum_{s,t=1}^n P_{s_i} P_{t_j} \delta_{st} = \sum_{s=1}^n P_{s_i} P_{s_j} \\
 &= (P^T P)_{ij}
 \end{aligned}$$

$$\implies \text{Vol}(P_{v_1, \dots, v_k}) = \sqrt{\det(P^T P)}$$

$$(\implies \text{In } \mathbb{R}_3, \text{Vol}(P_{a,b,c}) = \sqrt{\det(P^T P)} = \sqrt{(\det P)^2} = |\det P|)$$

**Thm** (Cauchy-Binet).

$$\det(U^T U) = \sum_{\substack{1 \leq i_1 < \dots < i_k \leq n \\ \binom{n}{k}}} U \begin{pmatrix} i_1 & \dots & i_k \\ 1 & \dots & k \end{pmatrix}^2$$

$$U^T U = \begin{pmatrix} U_{11} & \dots & U_{n1} \\ \vdots & & \vdots \\ U_{1k} & \dots & U_{nk} \\ \uparrow \\ i_1 & i_k \end{pmatrix} \begin{pmatrix} U_{11} & \dots & U_{1k} \\ \vdots & & \vdots & \leftarrow i_1 \ i_k \\ U_{n1} & \dots & U_{nk} \end{pmatrix}$$

**Corollary** (Generalized Pythagorean Theorem).

$$\text{Vol}(P_{v_1, \dots, v_k})^2 = \sum_{1 \leq i_1 < \dots < i_k \leq n} U \begin{pmatrix} i_1 & \dots & i_k \\ 1 & \dots & k \end{pmatrix}^2 \quad (k = 1, |V|^2 = \langle V, e_1 \rangle^2 + \dots + \langle V, e_n \rangle^2)$$

## 8.1 向量与坐标系

### 8.1.1 向量的定义与向量的加法和数乘

### 8.1.2 向量的共线和共面

### 8.1.3 向量的点乘和叉乘

### 8.1.4 向量的坐标表示

### 8.1.5 空间坐标系

## 8.2 Lines and Planes in $\mathbb{R}^3$

### 8.2.1 Plane

2-dim'l affine subspace.

- (General Equation)  $\Pi : ax + by + cz + d = 0$  **Implicit equation**

Suppose  $r_0 = (x_0, y_0, z_0) \in \Pi$ , i.e.  $ax_0 + by_0 + cz_0 + d = 0$

$$\begin{aligned} ax + by + cz + d = 0 &\iff a(x - x_0) + b(y - y_0) + c(z - z_0) = 0 \\ &\iff \langle \vec{r} - \vec{r}_0, \vec{n} \rangle = 0 \text{ where } \vec{n} = (a, b, c)^T \\ &\quad \text{(Point-Norm Form 点法式)} \end{aligned}$$

- (Point-Directions Form) Take  $\vec{r}_0 \in \Pi$ ,  $Q, R \in \Pi$ , s.t.  $\overrightarrow{PQ}, \overrightarrow{PR}$  linearly independent,  
 $\vec{u} = \overrightarrow{PQ}$ ,  $\vec{v} = \overrightarrow{PR}$ , then  $\Pi$  can be explicitly parameterized by

$$\vec{r} = \vec{r}_0 + s\vec{u} + t\vec{v}$$

$$\vec{n} = \overrightarrow{PQ} \times \overrightarrow{PR} \text{ normal vector to } \Pi$$

### 8.2.2 Line

1-dim'l affine subspace, i.e. shift of 1-dim'l vector subspace.

- (Point-Direction Form)  $\vec{r} = \vec{r}_0 + t\vec{v}$ ,  $t \in \mathbb{R}$  **Explicit equation 显式表达**
- (Plane-Intersection) The set of  $\mathbb{R}^3$  defined by  $\begin{cases} a_1x + b_1y + c_1z + d_1 = 0 \\ a_2x + b_2y + c_2z + d_2 = 0 \end{cases}$ , satisfying  
 $\text{rk} \begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{pmatrix} = 2$ . **Implicit equation 隐式表达**

- (1)  $\implies$  (2)  $(\vec{r} - \vec{r}_0) \times \vec{v} = \vec{0}$

$$\begin{aligned} \begin{pmatrix} i & j & k \\ x - x_0 & y - y_0 & z - z_0 \\ l & m & n \end{pmatrix} = 0 &\implies \begin{cases} n(y - y_0) = m(z - z_0) \\ l(z - z_0) = n(x - x_0) \\ m(x - x_0) = l(y - y_0) \end{cases} \\ &\implies \left( \begin{array}{ccc|c} 0 & n & -m & * \\ -n & 0 & l & * \\ m & l & 0 & * \end{array} \right) \end{aligned}$$

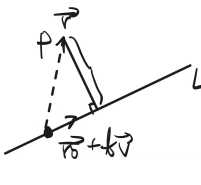
- (2)  $\implies$  (1)  $\xrightarrow{\text{Particular solution}} \vec{r}_0 + t(\vec{n}_1 \times \vec{n}_2)$  where  $\vec{n}_1 = (a_1, b_1, c_1)^T$ ,  $\vec{n}_2 = (a_2, b_2, c_2)^T$

### 8.2.3 Relation between Lines & Planes

1. Two planes: 
$$\begin{cases} \Pi_1 : a_1x + b_1y + c_1z + d_1 = 0 & \rightsquigarrow \vec{n}_1 \\ \Pi_2 : a_2x + b_2y + c_2z & \rightsquigarrow \vec{n}_2 \end{cases}$$

(a) Parallel:  $\vec{n}_1 \parallel \vec{n}_2$

(b) Intersecting: " $\theta = \arccos \frac{\langle \vec{n}_1, \vec{n}_2 \rangle}{|\vec{n}_1| |\vec{n}_2|}$ "

2. Point & line: 
$$d(P, L) = \frac{|(\vec{r} - \vec{r}_0) \times \vec{v}|}{|\vec{v}|}$$
 

Point & plane:  $\Pi : ax + by + cz + d = 0 \implies d(P, \Pi) = \frac{|ax + by + cz + d|}{\sqrt{a^2 + b^2 + c^2}}$

$$P = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

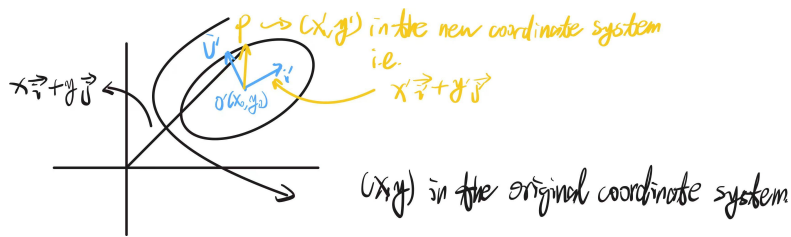
$$d(P, \Pi) = (\vec{r} - \vec{r}_0) \cdot \vec{n} = \frac{(a(x-x_0) + b(y-y_0) + c(z-z_0))}{\sqrt{a^2 + b^2 + c^2}}$$

3. Two lines: 
$$\begin{cases} L_1 : \vec{r} = \vec{r}_1 + t\vec{v}_1 \\ L_2 : \vec{r} = \vec{r}_2 + t\vec{v}_2 \end{cases}$$

(a) Co-planar  $\iff \langle \vec{r}_1 - \vec{r}_2, \vec{v}_1 \times \vec{v}_2 \rangle = 0$

(b) Distance:  $\left\langle \vec{r}_2 - \vec{r}_1, \frac{\vec{v}_1 \times \vec{v}_2}{|\vec{v}_1 \times \vec{v}_2|} \right\rangle$

### 8.3 Conic Section/Curve & Quadric Surface



$$\Rightarrow \begin{pmatrix} \vec{i} & \vec{j} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \overrightarrow{OP} = \overrightarrow{OO'} + \overrightarrow{O'P} \\ = \begin{pmatrix} \vec{i}' & \vec{j}' \end{pmatrix} \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} + \begin{pmatrix} \vec{i}' & \vec{j}' \end{pmatrix} \begin{pmatrix} x' \\ y' \end{pmatrix} \\ = \begin{pmatrix} \vec{i}' & \vec{j}' \end{pmatrix} \begin{pmatrix} x' \\ y' \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = P \begin{pmatrix} x' \\ y' \end{pmatrix} + \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} \text{ coordinate transformation}$$

#### Quadric in $\mathbb{R}^n$

$$f(x_1, \dots, x_n) = \sum_{i,j=1}^n x_i A_{ij} x_j + \sum_{i=1}^n b_i x_i + C, \text{ where } A = A^T \\ = X^T A X + b^T X + C, \text{ where } X = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

Question: Can we find coordinate transformation  $x = Py$  where  $P \in O(n)$  to simplify the expression:

$$y^T P^T A P y + b^T P y + C$$

1. Take  $P$  s.t.

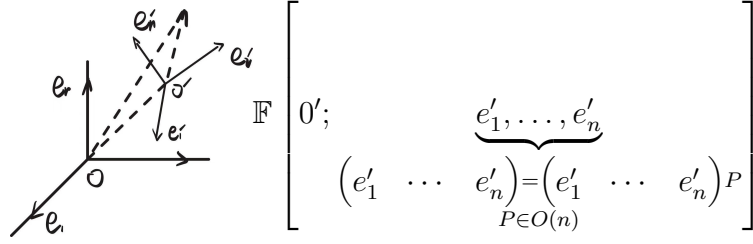
$$P^T A P = \begin{pmatrix} \lambda_1 & & & & & & & & \\ & \ddots & & & & & & & \\ & & \lambda_{r_+} & & & & & & \\ & & & -\lambda_{r_++1} & & & & & \\ & & & & \ddots & & & & \\ & & & & & \lambda_{r_++r_-} & & & \\ & & & & & & 0 & & \\ & & & & & & & \ddots & \\ & & & & & & & & 0 \end{pmatrix}$$

2.  $y = z + \begin{pmatrix} \vdots \end{pmatrix}$

$$3. \lambda_1 z_1^2 + \cdots + \lambda_{r_+} z_{r_+}^2 - \left( \lambda_{r_++1} z_{r_++1}^2 + \cdots \right)$$

$$\text{Quadric surface: } \left\{ \vec{x} \in \mathbb{R}^n \mid f(\vec{x}) = \vec{x}^T A \vec{x} + \vec{b}^T \vec{x} + c = 1 \right\}, A = A^T$$

## Descartes / Catesian Coordinate



A point has coordinate

$$\left\{ \begin{array}{l} \vec{x}' = \begin{pmatrix} x'_1 \\ \vdots \\ x'_n \end{pmatrix} \text{ in } \mathbb{F}[0'; e'_1, \dots, e'_n] \\ \vec{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \text{ in } \mathbb{F}[0; e_1, \dots, e_n] \end{array} \right. \Rightarrow \begin{array}{l} \vec{OO'} \\ = (e_1 \dots e_n) \vec{v} \end{array} + \begin{pmatrix} e'_1 & \dots & e'_n \end{pmatrix} \vec{x}' = \begin{pmatrix} e_1 & \dots & e_n \end{pmatrix} \vec{x}$$

$$\Rightarrow \vec{x} = \vec{v} + p \vec{x}'$$

Step 1.  $\vec{x} = p \vec{x}' \Rightarrow x' P^T A P x' + b^T P x' + c = 0$   
 对角阵

Find  $P$  s.t.

$$\xrightarrow{\text{orthogonal diagonalization}} P^T A P = \Lambda = \text{diag} \left( \lambda_1, \dots, \lambda_{r_+}, -\lambda_{r_++1}, \dots, -\lambda_{r_++r_-}, \underbrace{0, \dots, 0}_{n-\text{rank } A} \right)$$

$$\text{then } \lambda_1 x_1'^2 + \dots + \lambda_{r_+} x_{r_+}'^2 - \lambda_{r_++1} x_{r_++1}'^2 - \dots - \lambda_{r_++r_-} x_{r_++r_-}'^2 + \mathbf{b}'_1 \mathbf{x}'_1 + \dots + \mathbf{b}'_n \mathbf{x}'_n + c = 0$$

Step 2. Completing square for the variables  $x'_1, \dots, x'_{r_+}, x'_{r_++1}, \dots, x'_{r_++r_-}$

$$\begin{pmatrix} x'_1 \\ \vdots \\ x'_{r_+} \\ x'_{r_++1} \\ \vdots \\ x'_{r_++r_-} \\ \vdots \\ x'_n \end{pmatrix} = \begin{pmatrix} x''_1 - \frac{b'_1}{2\lambda_1} \\ \vdots \\ x''_{r_+} - \frac{b'_{r_+}}{2\lambda_{r_+}} \\ x''_{r_++1} - \frac{b'_{r_++1}}{2\lambda_{r_++1}} \\ \vdots \\ x''_{r_++r_-} - \frac{b'_{r_++r_-}}{2\lambda_{r_++r_-}} \\ x''_{r_++r_-+1} \\ \vdots \\ x''_n \end{pmatrix}$$

$$\text{i.e. } x'_1 = x'' + b''$$

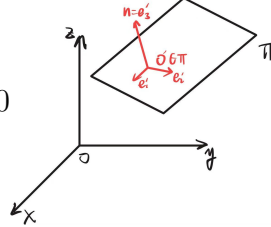
$$\lambda_1 x_1''^2 + \dots + \lambda_{r_+} x_{r_+}''^2 - \lambda_{r_++1} x_{r_++1}''^2 - \dots - \lambda_{r_++r_-} x_{r_++r_-}''^2 + b'_{r_++r_-+1} x_{r_++r_-+1}'' + \dots + b'_n x_n'' + c = 0$$

Step 3. Case 1.

$$b'_{r+1} = \dots = b'_n = 0 \rightsquigarrow x''^T \text{diag}(\lambda_1, \dots, \lambda_{r_+}, -\lambda_{r_++1}, \dots, -\lambda_{r_++r_-}, 0, \dots, 0) x'' + c = 0$$

Case 2. We can do another orthogonal transformation on the variable  $x''_{r+1}, \dots, x''_n$  to simplify the 1st-order term s.t. the 1st order term becomes  $b'''_{r+1} x'''_{r+1}$

Back to plane in  $\mathbb{R}^3$ :  $ax + by + cz + d = 0$



Let  $e'_3 = n = \frac{(a,b,c)}{\sqrt{a^2+b^2+c^2}}$ ,  $e'_1, e'_2$  parallel to  $\Pi$  s.t.  $(e'_1, e'_2, n)$  is an orthonormal basis, then in the Cartesian coordinate  $\mathbb{F}[0'; e'_1, e'_2, e'_3]$ ,  $\Pi$  is defined by  $z' = 0$

$$\begin{pmatrix} a & b & c \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} + d = 0$$

Let  $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = P \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} + q$  (Euclidean transformation), then the equation is changed to

$$\begin{pmatrix} a & b & c \end{pmatrix} P \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} + \begin{pmatrix} a & b & c \end{pmatrix} q + d = 0$$

find  $P \in O(3)$ ,  $q \in \mathbb{R}^3$  s.t.  $\begin{pmatrix} a & b & c \end{pmatrix} P = \begin{pmatrix} 0 & 0 & c \end{pmatrix}$ ,  $c \neq 0$ ,  $\begin{pmatrix} a & b & c \end{pmatrix} q + d = 0$

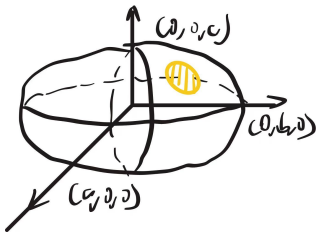
**Thm.** Given  $S = \{ \vec{x} \in \mathbb{R}^n \mid f(\vec{x}) = \vec{x}^T A \vec{x} + \vec{b}^T \vec{x} + c = 0 \}$ , we can find a coordinate transformation of the form  $\vec{x} = P \vec{y} + \vec{d}$ , where  $P \in SO(n) = \{ Q \in O(n) \mid \det Q = 1 \}$  and  $\vec{d} \in \mathbb{R}^n$ , such that the equation  $f(\vec{x}) = 0$  becomes

$$(1) \sum_{i=1}^{r_+} \lambda_i y_i^2 - \sum_{j=r_++1}^{r_++r_-} \lambda_j y_j^2 = c \text{ or}$$

$$(2) \sum_{i=1}^{r_+} \lambda_i y_i^2 - \sum_{j=r_++1}^{r_++r_-} \lambda_j y_j^2 - y_{r_++r_-+1} = 0$$

## Classification of Quadric Surface in $\mathbb{R}^3$

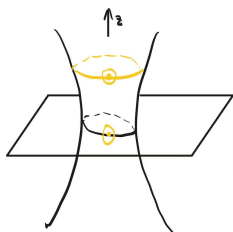
### 1. rank $A = 3$

$$(+, +, +; +) \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$


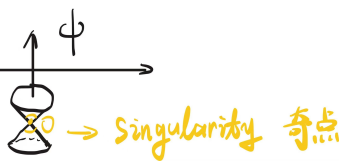
ellipsoid

$$(+, +, +; 0) \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 0$$

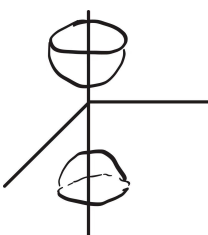
$$(+, +, +; -) \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = -1$$

$$(+, +, -; +) \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$


single sheeted hyperboloid 单叶双曲面

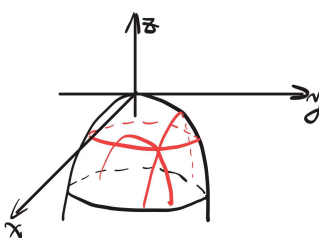
$$(+, +, -; 0) \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0$$


Singularity 奇点

$$(+, +, -; -) \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = -1$$


Double sheeted hyperboloid 双叶双曲面

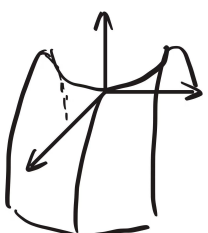
### 2. rank $A = 2$

$$(+, +, 0; +; 0) \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} + z = 0$$


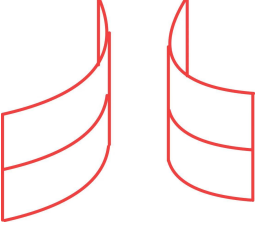
Elliptical paraboloid 椭圆抛物面

$$(+, +, 0; 0; +) \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

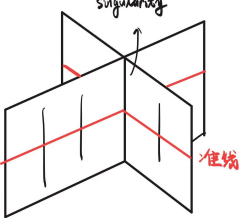

Elliptical cylinder 椭圆柱面

$$(+, -, 0; +; 0) \quad \frac{x^2}{a^2} - \frac{y^2}{b^2} + z = 0$$


Hyperbolic paraboloid/  
Saddle surface 鞍面

$$\begin{cases} (+, -, 0; 0; +) & \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \\ (+, -, 0; 0; -) & \frac{x^2}{a^2} - \frac{y^2}{b^2} = -1 \end{cases}$$


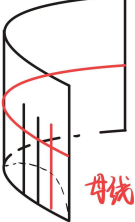
Hyperbolic cylinder

$$(+, -, 0; -; 0) \quad \frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$$


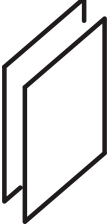
singularity

准线

3. rank  $A = 1$

$$(+, 0, 0; +; 0) \quad \frac{x^2}{a^2} + y = 0$$


母线

$$(+, 0, 0; 0; +)$$


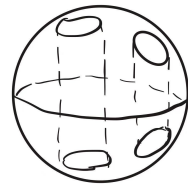
Double plane

$$(+, 0, 0; 0; 0)$$

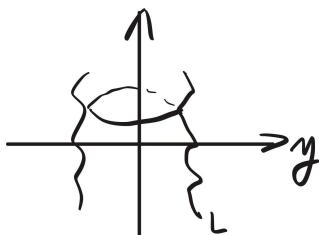
$$(+, 0, 0; 0; -) \rightarrow \emptyset$$

## Intersection of Quadric Surface and Plane

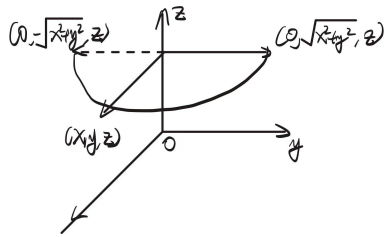
$$\begin{cases} f(\vec{x}) = \vec{x}^T A \vec{x} + \vec{b}^T \vec{x} + c = 0 \\ \vec{n}^T \vec{x} + d = 0 \end{cases} \quad \begin{cases} x^2 + y^2 + z^2 = 1 \\ (x - a)^2 + y^2 = r^2 \end{cases}$$



## Surface of Revolution 旋转



$F(y, z) = 0 \rightarrow$  curve  $L$  revolved along  $Z$  axis  $\Rightarrow$  Surface implicit equation?

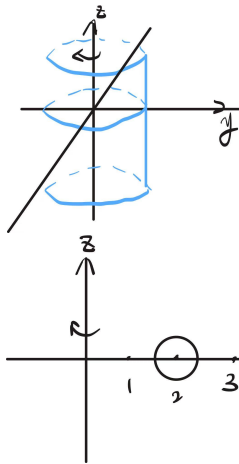


$$\Rightarrow F\left(\pm\sqrt{x^2+y^2}, z\right)=0$$

$\Leftrightarrow (x, y, z)$  lies on the surface of revolution

**Eg.**

$$L=\left\{\left(0, y, z\right) \mid y=z\right\}$$

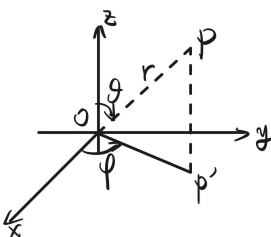


$$\begin{aligned} \pm\sqrt{x^2+y^2}=z &\rightsquigarrow x^2+y^2-z^2=0 \\ x^2+y^2+4\pm 4\sqrt{x^2+y^2}+z^2=1 &\Rightarrow\left(x^2+y^2+z^2+3\right)^2=16\left(x^2+y^2\right) \\ (y-2)^2+z^2=1 &\rightsquigarrow\left(\pm\sqrt{x^2+y^2}-2\right)^2+z^2=1 \end{aligned}$$

## 8.4 坐标变换和其他常用坐标系

### 8.4.1 坐标变换

### 8.4.2 Other Coordinate System

1.   $(r, \theta, \varphi) \begin{cases} r \in [0, +\infty) \\ \theta \in [0, \pi] \\ \varphi \in [0, 2\pi) \end{cases} \rightsquigarrow (x, y, z)^T = (r \sin \theta \cos \varphi, r \sin \theta \sin \varphi, r \cos \theta)$

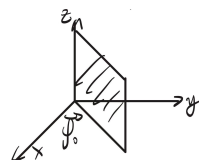
1:1 on the “open point”

$$x^2 + y^2 = r^2 \sin^2 \theta \quad z = r \cos \theta$$

$$\begin{cases} r = \sqrt{x^2 + y^2 + z^2} \\ \theta = \arctan \frac{\sqrt{x^2 + y^2}}{z} \\ \varphi = \arctan \frac{y}{x} \end{cases}$$

•  $r = r_0$  

•  $\theta = \theta_0$  

•  $\varphi = \varphi_0$  

Curvilinear coordinate system

### 2. Cylindrical coordinate $(r, \theta, z)$

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \\ z = z \end{cases} \iff \begin{cases} r = \sqrt{x^2 + y^2} \\ \theta = \arctan \frac{y}{x} \\ z = z \end{cases}$$

## 9 多变量函数的微分学

### 9.1 多变量函数及其连续性

#### 9.1.1 Point-set

Point-set  $\mathbb{R}^n, n \geq 2$

Recall: Distance on  $\mathbb{R}^n$  (Euclidean distance)

$$d : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R} \\ (x, y) \mapsto |x - y|$$

- Nonnegativity:  $d(x, y) \geq 0, \forall x, y \in \mathbb{R}^n$ ; “=” is achieved iff  $x = y$
- Symmetric:  $d(x, y) = d(y, x)$
- Triangle inequality:  $d(x, y) \leq d(x, z) + d(z, y), \forall x, y, z \in \mathbb{R}^n$

$$|u + v| \leq |u| + |v| \iff \langle u + v, u + v \rangle \leq \langle u, u \rangle + 2|u||v| + \langle v, v \rangle \\ \iff \langle u, v \rangle \leq |u||v| \text{ (Cauchy-Schwarz inequality)}$$

**Def.** The disc  $B(M_0, r) = \{M \in \mathbb{R}^n \mid d(M, M_0) < r\}$  is called a disc-neighborhood of  $M_0$ , which is also denoted by  $B_r(M_0)$ .  $B^-(M_0, r) = B(M_0, r) \setminus \{M_0\}$  is called punctured disc.

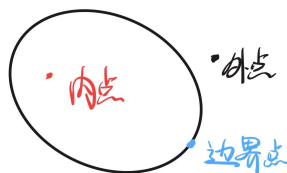
**Def.**  $E \subseteq \mathbb{R}^n$  is called bounded if  $E \subseteq B(0, R)$  for some  $R > 0$ , and the number

$$\text{diam } E \triangleq \sup \{d(M', M'') \mid M', M'' \in E\}$$

is called the diameter of  $E$ .

**Def.** Let  $E \subseteq \mathbb{R}^n$  and  $E^C = \mathbb{R}^n \setminus E$  be its complement.

1. If  $\exists r > 0$ , s.t.  $B(M, r) \subseteq E$ , then  $M$  is called an interior point 内点 of  $E$ .
2. If  $\exists r > 0$ , s.t.  $B(M, r) \subseteq E^C$ , then  $M$  is called an exterior point 外点 of  $E$ .
3. If  $\exists r > 0, B(M, r) \cap E \neq \emptyset, B(M, r) \cap E^C \neq \emptyset$ , then  $M$  is called a boundary point 边界点 of  $E$ .



4.  $\overset{\circ}{E} = \{M \in \mathbb{R}^n \mid M \text{ is an interior point of } E\}$ , is called the interior 内部 of  $E$ .
5.  **$\partial$ : partial**  $\partial E = \{M \in \mathbb{R}^n \mid M \text{ is a boundary point of } E\}$ , is called the boundary of  $E$ .
6. If  $\exists r > 0$ , s.t.  $B(M, r) \cap E = \{M\}$ , then  $M$  is called an isolated point 孤立点.
7. If  $\forall r > 0$ ,  $B^-(M, r) \cap E \neq \emptyset$ , then  $M$  is called a accumulation point 聚点.  
 $\overline{E} = E \cup \partial E$ , is called the closure 闭包 of  $E$ .

**Rmk.** 1. Interior point implies accumulation point

(a)  $M \in E \implies M$  is an accumulation point

(b)  $M \notin E \implies$  any punctured disc neigh. contains points in  $E \implies$  accumulation

2. boundary point  $\implies$  either isolated point or accumulation point

boundary point + non-isolated  $\implies$  accumulation point

**Def.** Let  $\{M_n\}$  be a sequence of points on  $\mathbb{R}^2$ , and  $M_0 \in \mathbb{R}^2$ . If  $\lim_{n \rightarrow +\infty} d(M_n, M_0) = 0$ , we say  $M_n$  converges to  $M_0$  (as  $n \rightarrow +\infty$ ), also denoted by  $\lim_{n \rightarrow +\infty} M_n = M_0$

$M_n = \begin{pmatrix} x_n \\ y_n \end{pmatrix}$ ,  $M_0 = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$ , then  $\lim_{n \rightarrow +\infty} M_n = M_0 \iff \lim_{n \rightarrow +\infty} x_n = x_0$ , and  $\lim_{n \rightarrow +\infty} y_n = y_0$   
i.e.  $\sqrt{(x_n - x_0)^2 + (y_n - y_0)^2} \rightarrow 0$ ,  $|x_n - x_0|, |y_n - y_0| \leq d(M_n, M_0) \leq |x_n - x_0| + |y_n - y_0|$

A sequence of points  $\{M_n\}$  is called a Cauchy Seq. if

$$\forall \varepsilon > 0, \exists N = N(\varepsilon) \in \mathbb{N}_+, \text{ s.t. } \forall n, m > N, d(M_n, M_m) < \varepsilon$$

**Thm.** 1. (Completeness) All Cauchy seq. in  $\mathbb{R}^n$  has convergent subseq.

2. (Bolzano-Weierstrass Thm.) All bounded seq. in  $\mathbb{R}^n$  has convergent subseq.

**Def** (Open set / Closed set). If  $E = \overset{\circ}{E}$ , we say  $E$  is open. If  $E^C$  is open, we say  $E$  is closed.

**Eg.** Disc-neighborhood  $B(M, r) = \{O \in \mathbb{R}^n \mid d(P, M) < r\}$  is open.

$$\forall P \in B(M, r), d(P, M) < r$$

$$\implies B(P, r - d(P, M)) \subseteq B(M, r)$$

$$(d(Q, M) \leq d(Q, P) + d(P, M) < (r - d(P, M)) + d(P, M)) = r, \forall Q \in B(P, r - d(P, M))$$

$$\implies B(P, r) \text{ is open, } B(P, r)^C \text{ closed.}$$

$$\partial B(M, r) = \left\{ P \in \mathbb{R}^n \mid \boxed{d(P, M) = r} \right\} = S(M, r) \leftarrow \text{Sphere}$$

**Prop.** 1.  $E_1, E_2$  open  $\implies E_1 \cap E_2$  is open.

(If  $E_1, \dots, E_n$  open, then  $E_1 \cap \dots \cap E_n$  is open)

If  $\{E_\alpha\}_{\alpha \in \Lambda}$  is a family 族 of open sets, then  $\bigcup_{\alpha \in \Lambda} E_\alpha$  is open.

$$\begin{aligned} \Longleftarrow \forall x \in \bigcup_{\alpha \in \Lambda} E_\alpha &\implies x \in E_\alpha \text{ for some } \alpha \in \Lambda \\ &\implies \exists r > 0 \text{ s.t. } B(x, r) \subseteq E_\alpha \\ &\implies B(x, r) \subseteq \bigcup_{\alpha \in \Lambda} E_\alpha \end{aligned}$$

$$\bigcap_{n=1}^{\infty} B\left(0, \frac{1}{n}\right) = \{0\} \text{ not open}$$

2.  $E_1, E_2$  closed  $\implies E_1 \cup E_2$  is closed.

(If  $E_1, \dots, E_n$  closed, then  $E_1 \cup \dots \cup E_n$  is closed)

If  $\{E_\alpha\}_{\alpha \in \Lambda}$  is a family of closed sets, then  $\bigcap_{\alpha \in \Lambda} E_\alpha$  is closed.

3.  $E$  is open iff  $E \cap \partial E = \emptyset$ , i.e.  $\partial E \subseteq E^C$

4.  $E$  is closed iff  $E^C \cap \partial E = \emptyset$ , i.e.  $\partial E \subseteq E$   
 $\Downarrow$   
 $E^C \text{ open} \implies E^C \cap \partial(E^C) = E^C \cap \partial E = \emptyset$

5.  $E$  is closed iff it contains all of its accumulation points.

**Pf.** 1.  $\forall x \in E_1 \cap E_2, \exists r_1, r_2 > 0$ , s.t.  $B(x, r_1) \subseteq E_1, B(x, r_2) \subseteq E_2$

$$\implies B(x, \min\{r_1, r_2\}) \subseteq E_1 \cap E_2$$

$$\begin{aligned} 2. (E_1 \cup E_2)^C &= E_1^C \cap E_2^C \\ (\bigcap_{\alpha \in \Lambda} E_\alpha)^C &= \bigcup_{\alpha \in \Lambda} E_\alpha^C \end{aligned}$$

$$\begin{aligned} x \in \left(\bigcap_{\alpha \in \Lambda} E_\alpha\right)^C &\iff x \notin \bigcap_{\alpha \in \Lambda} E_\alpha \\ &\iff x \notin E_\alpha \text{ for some } \alpha \in \Lambda \\ &\iff x \in E_\alpha^C \text{ for some } \alpha \in \Lambda \text{ i.e. } x \in \bigcup_{\alpha \in \Lambda} E_\alpha^C \end{aligned}$$

3. “ $\implies$ ”  $E$  is open  $\implies \forall x \in E, \exists r > 0$  s.t.  $B(x, r) \subseteq E \implies x \notin \partial E \implies \partial E \subset E^C$

“ $\Leftarrow$ ” If  $\partial E \cap E = \emptyset$ , then  $\forall x \in E \implies x \notin \partial E \xrightarrow{\text{反证}} \exists r > 0$  s.t.  $B(x, r) \subseteq E \implies x$  is an interior  $\implies E$  open.

5. " $\implies$ "  $E$  closed,  $\boxed{x \text{ is an accumulation point}}$ , we want to prove  $x \in E$  by contradiction.

Suppose  $x \notin E$ , then  $x \in E^C$  which is open. Then  $B(x, r) \subseteq E^C$  for some  $r > 0$ , so  $x$  is not accumulation point of  $E$ , it's a contradiction.

Therefore,  $x \in E$ .

" $\Leftarrow$ "  $\forall x \in E^C$ , then  $x$  is not accumulation point by assumption.

$$\implies \exists r > 0 \text{ s.t. } B^-(x, r) \cap E = \emptyset \implies B^-(x, r) \subseteq E^C$$

$$\implies B(x, r) \subseteq E^C \implies E^C \text{ open} \implies E \text{ closed.}$$

**Def** (Connectivity). A subset  $E \subseteq \mathbb{R}^n$  is called connected if for any nontrivial  $(A, B \neq \emptyset)$  decomposition  $E = \underset{\text{disjoint set}}{A \sqcup B}$ , either  $A$  contains some accumulation points of  $B$

$$x \in A \cap \overline{B} \neq \emptyset \implies x \in \overline{B} \setminus B$$

$$\implies x \in (\partial B) \cap B^C$$

$$\implies x \text{ is accumulation of } B, B \cap \overline{A} \neq \emptyset$$

**Def** (Path-connected 道路连通). If  $\forall P, Q \in E, \exists$  continuous map

$$\gamma : [0, 1] \rightarrow \mathbb{R}^n$$

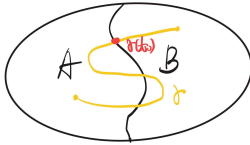
$$t \mapsto \begin{pmatrix} x_1(t) \\ \vdots \\ x_n(t) \end{pmatrix}$$

s.t.  $\gamma([0, 1]) \subseteq E$ , and  $\gamma(0) = P, \gamma(1) = Q$

**Eg.**  $E = \{(0, y) \mid -1 \leq y \leq 1\} \cup \{(x, \sin \frac{1}{x}) \mid x \in (0, 1]\}$  **Not path connected, but it is connected**

**Thm.** If  $E$  is path-connected, then  $E$  is connected.

**Pf.** Argue by contradiction. Suppose  $E$  is not connected / disconnected. Then  $A, B \neq \emptyset, A \cap B = \emptyset$ , s.t.  $A \sqcup B = E$  and  $A \cap \overline{B} = \emptyset, B \cap \overline{A} = \emptyset$ . By path-connectedness, take  $x \in A, y \in B, \gamma : [0, 1] \rightarrow \mathbb{R}^n$ , path connecting  $x$  and  $y$  in  $E$  to  $\triangleq \sup \{t \in [0, 1] \mid \gamma(t) \in A\} \neq \emptyset$



1.  $r(t_0) \in A \implies t_0 \neq 1, \forall t > t_0, r(t) \in B$   
 $\implies r(t_0)$  is accumulation point of  $B$   
 $\implies r(t_0) \in \overline{B} \subseteq A^C$ , contradiction.

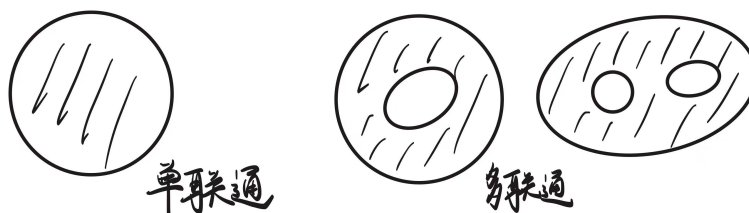
2.  $r(t_0) \in B \implies \exists t_n \nearrow t_0$  s.t.  $r(t_n) \in A$   
 $\implies r(t_0)$  is accumulation point of  $A$   
 $\implies r(t_0) \in \overline{A} \subseteq B^C$ , contradiction.

**Rmk.** Open set is connected iff it is path-connected.

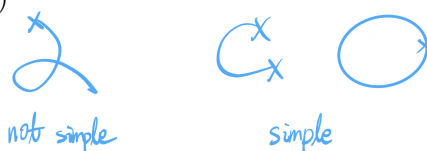
(Pick any  $P \in X$ , define

$$U = \{Q \in X \mid Q \text{ and } P \text{ can be connected by a path in } X\},$$

prove  $U = X$ )



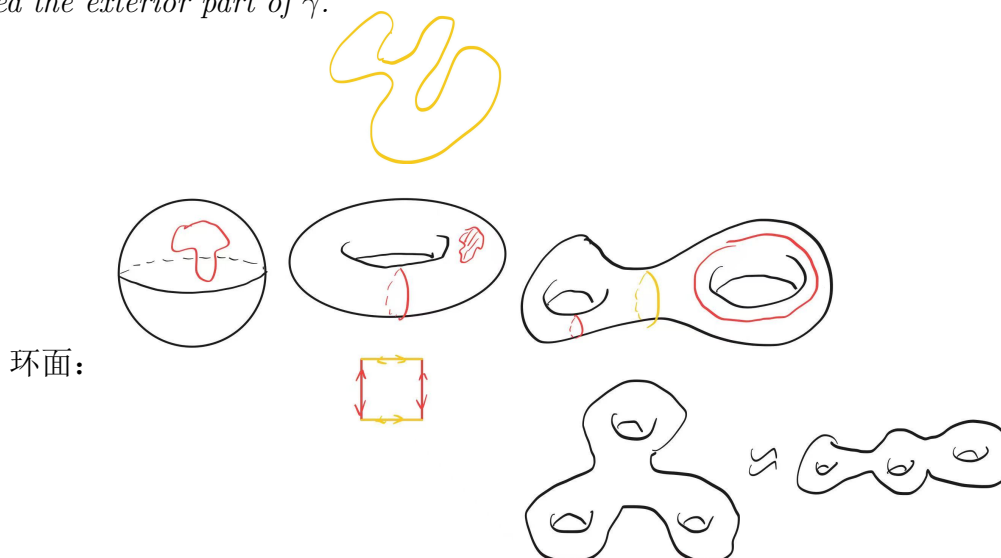
**Def** (Single Curve). If a curve:  $\gamma : [\alpha, \beta] \rightarrow \mathbb{R}^2$  satisfies  $\gamma(t_1) \neq \gamma(t_2)$  for any  $\alpha \leq t_1 < t_2 \leq \beta$ , then  $\gamma$  is called a single curve. (Not self-intersection)



**Def** (Loop / Closed Curve). If the curve  $\gamma : [\alpha, \beta] \rightarrow \mathbb{R}^2$  has the same starting and ending point, i.e.  $\gamma(\alpha) = \gamma(\beta)$ , then  $\gamma$  is called a loop, or a closed curve.



**Thm** (Jordan Curve Theorem). Any simple closed curve  $\gamma$  on  $\mathbb{R}^2$  divides  $\mathbb{R}^2$  into two open connected sets, one is bounded called the interior part of  $\gamma$ , and the other which is unbounded called the exterior part of  $\gamma$ .



If the interior point of any single closed curve in  $D$  (connected open set) is contained in  $D$ , then  $D$  is called **simply-connected**. Otherwise, it is called multi-connected.

**Rmk.** *Topological definition (depends on the crucial notion homotopy 同伦)*

*For  $\mathbb{R}^2$ , any loop is null-homotopic 零伦, i.e. it is homotopic to a trivial loop.*

$\Pi_1(x)$  基本群

### 9.1.2 Multi-variables Function

Any map

$$f : D (\subseteq \mathbb{R}^2) \rightarrow \mathbb{R}$$

$$\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \mapsto z = f(x_1, \dots, x_n)$$

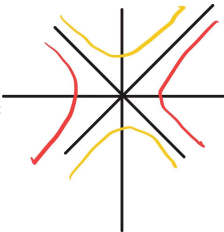
Cartesian Coordinate

is called a multi-variables function.

1. Explicit formula:  $z = f(x_1, \dots, x_n)$
2. Graph:  $\text{Graph}(f) = \{(x_1, \dots, x_n), f(x_1, \dots, x_n) \mid (x_1, \dots, x_n) \in D\} \subseteq \mathbb{R}^{n+1} \rightarrow$ 
  - $z = x^2 - y^2$  马鞍面
  - $\{x^2 + y^2 + z^2 = 1, z \geq 0\}$  半球:  $z = \sqrt{1 - x^2 - y^2}, D = \{x^2 + y^2 \leq 1\}$
3. Level set representation

$$\forall c \subseteq \mathbb{R}, f^{-1}(c) = \{(x_1, \dots, x_n) \in D \mid f(x_1, \dots, x_n) = c\}$$

is called the level set at the level  $c$ .

**Eg.** •  $z = f(x, y) = x^2 - y^2 \rightarrow f^{-1}(c) =$  

•  $w = f(x, y, z) = x^2 + y^2 - z^2$



### Vector-valued Functions

$$f : D (\subseteq \mathbb{R}^n) \rightarrow \mathbb{R}^m$$

$$(x_1, \dots, x_n) \mapsto (y_1, \dots, y_m) = (f_1(x_1, \dots, x_n), \dots, f_m(x_1, \dots, x_n))$$

**Eg.** 1.  $n = 1, m = 3$  parameterized curve

$$\vec{r} : D (\subseteq \mathbb{R}) \rightarrow \mathbb{R}^3$$

$$t \mapsto (x(t), y(t), z(t))$$

2.  $n = 2, m = 3$  parameterized surface in  $\mathbb{R}^3$

$$\begin{aligned}\vec{r} : D (\subseteq \mathbb{R}^2) &\rightarrow \mathbb{R}^3 \\ (u, v) &\mapsto (x(u, v), y(u, v), z(u, v))\end{aligned}$$

Eg.

$$\begin{aligned}\vec{r} : [0, \pi] \times [0, 2\pi] &\rightarrow \mathbb{R}^3 \\ (\theta, \varphi) &\mapsto (\sin \theta \cos \varphi, \sin \theta \sin \varphi, \cos \theta)\end{aligned}$$

is the spherical coordinate parameterization of the  $S^2 = \partial B(0, 1)$

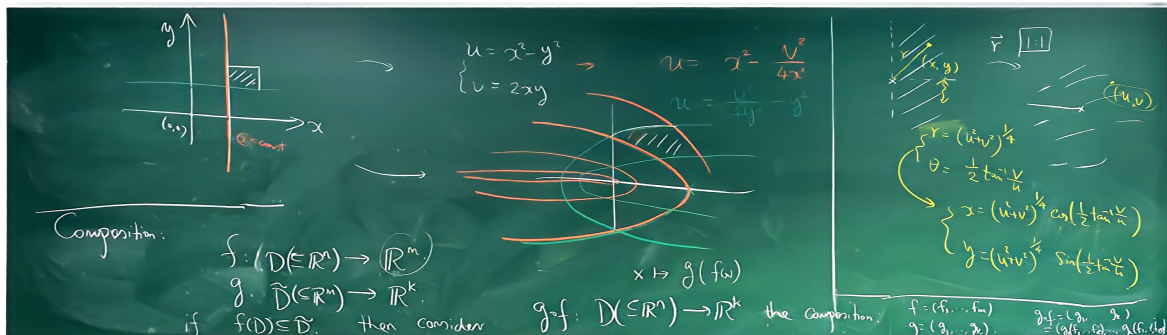
3.  $n = m$  a one-to-one vector-valued function  $f : D (\subseteq \mathbb{R}^b) \rightarrow \tilde{D} (\subseteq \mathbb{R}^b)$  is called a coordinate.

## Transformation

Eg.

$$\begin{aligned}\vec{r} : \mathbb{R}^2 &\rightarrow \mathbb{R}^2 \\ (x, y) &\mapsto (x^2 - y^2, 2xy) \\ (r, \theta) &= (r^2 \cos 2\theta, r^2 \sin 2\theta) \rightarrow (r^2, 2\theta)\end{aligned}$$

$$\begin{aligned}x + iy &= z \\ w = u + iv &= z^2 \\ \text{Conformal 共形}\end{aligned}$$



### 9.1.3 Limit

**Def (Limit).** Let  $f : D (\subseteq \mathbb{R}^2) \rightarrow \mathbb{R}$ ,  $M_0$  is an accumulation point of  $D$ . If for some  $a \in \mathbb{R}$ ,  $\forall \varepsilon > 0$ ,  $\exists \delta = \delta(\varepsilon) > 0$  s.t.  $\forall M \in D \cap B^-(M_0, \delta)$ ,  $|f(M) - a| < \varepsilon$ , then we say  $f$  converges to  $a$  as  $M$  converges to  $M_0$ , simply denote it by  $\lim_{M \rightarrow M_0} f(M) = a$ .

$\forall \varepsilon > 0$ ,  $\exists \delta = \delta(\varepsilon) > 0$ , s.t.  $\forall (x, y) \in D$ ,  $(x, y) \neq M_0 = (x_0, y_0)$ ,  $|x - x_0| < \delta$ ,  $|y - y_0| < \delta$ ,  $|f(x, y) - a| < \varepsilon$

$$\left( \begin{array}{c} \lim_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} f(x, y) \\ \downarrow \\ \lim_{y \rightarrow y_0} \lim_{x \rightarrow x_0} f(x, y) \text{ 累次极限} \\ \lim_{M_0 + tv} f(M_0 + tv) \end{array} \right)$$

Similarly, we can define  $\lim_{\substack{x \rightarrow \infty \\ y \rightarrow \infty}} f(x, y)$ ,  $\lim_{\substack{x \rightarrow x_0 \\ y \rightarrow \infty}} f(x, y)$ ,  $\dots$

**Eg.** 1. Prove  $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{x^4 + y^2} = 0$

**Pf.**  $\left| \frac{x^2 y^2}{x^4 + y^2} \right| = |y| \left| \frac{x^2 y}{x^4 + y^2} \right| \leq \frac{1}{2} |y|$

$\forall \varepsilon > 0$ ,  $\exists \delta = \varepsilon$  s.t.  $\forall (x, y) \in \mathbb{R}^2 \setminus \{(0, 0)\}$ ,  $|x| < \delta$ ,  $|y| < \delta$

$$\left| \frac{x^2 y^2}{x^4 + y^2} \right| \leq \frac{1}{2} |y| < \varepsilon$$

*Q.E.D.*

2. Prove  $\lim_{(x,y) \rightarrow (0,0)} \frac{x^3 + y^3}{x^2 + y^2} = 0$  ( $\left| \frac{x^3 + y^3}{x^2 + y^2} = 0 \right| \leq |x| + |y|$ )

3. Consider  $f(x, y) = \frac{x^2 y}{x^4 + y^2}$ ,  $(x, y) \neq (0, 0)$

$$\lim_{y \rightarrow 0} \lim_{x \rightarrow 0} \frac{x^2 y}{x^4 + y^2} = 0 \quad \lim_{x \rightarrow 0} \lim_{y \rightarrow 0} \frac{x^2 y}{x^4 + y^2} = 0$$

$$f(x, y) = f(x, kx) = \frac{x^2 \cdot kx}{x^4 + k^2 x^2} = \frac{k}{k^2 + x^2} x \rightarrow 0, \text{ as } x \rightarrow 0$$

**If we consider  $y = kx^2$ ,  $f(x, y) = f(x, kx^2) = \frac{x^2 \cdot kx^2}{x^4 + k^2 x^4} = \frac{k}{1 + k^2} \rightarrow \frac{k}{1 + k^2} (x \rightarrow 0)$**

**$\implies \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^4 + y^2}$  does not exist**

**Thm.** Suppose  $f$  is defined on  $B^-(M_0, r)$ , then  $\lim_{M \rightarrow M_0} f(M) = a \iff$  for any continuous curve  $\gamma : (0, \delta) \rightarrow B^-(M_0, r)$  with  $\lim_{t \rightarrow 0^+} \gamma(t) = M_0$ , we have  $\lim_{t \rightarrow 0} f(\gamma(t)) = a$

**Pf.** “ $\implies$ ”  $\forall \varepsilon > 0$ ,  $\exists \delta > 0$  s.t.  $\forall M \in B^-(M_0, \delta)$ ,  $|f(M) - a| < \varepsilon$ . Then for any  $\gamma : (0, \delta) \rightarrow B^-(M_0, r)$  with  $\gamma(t) \rightarrow M_0 (t \rightarrow 0)$ .  $\exists \eta > 0$  s.t.  $\forall 0 < t < \eta$ ,  $\gamma(t) \in B^-(M_0, \delta)$ , then  $|f(\gamma(t))| < \varepsilon$

“ $\impliedby$ ” Argue by contradiction. Suppose  $\lim_{M \rightarrow M_0} f(M) \neq a$ , then  $\exists \varepsilon_0 > 0$  and  $\{M_n\} \subseteq B^-(M_0, r)$  s.t.  $M_n \rightarrow M_0 (n \rightarrow +\infty)$ , and  $|f(M_n) - a| \geq \varepsilon_0$

**Construct a  $\gamma : (0, 1) \rightarrow B^-(M_0, r)$ ,  $\gamma(\frac{1}{2^n})$  and  $\gamma(t) \rightarrow M_0 (t \rightarrow 0)$ , but  $f(\gamma(\frac{1}{2^n})) = f(M_n) \not\rightarrow a$ . Contradiction.**

**Prop** (Algebraic Rules). If  $\lim_{M \rightarrow M_0} f(M)$ ,  $\lim_{M \rightarrow M_0} g(M)$  both exists, then  $\lim_{M \rightarrow M_0} (f \pm g)(M)$ ,  $\lim_{M \rightarrow M_0} (f \cdot g)(M)$  exists and  $\lim_{M \rightarrow M_0} (\frac{f}{g})(M)$  exists provided  $\lim_{M \rightarrow M_0} g(M) \neq 0$ .

**Prop** (Heine Principle). Suppose  $M_0$  is an accumulation point of  $D$ , then  $\lim_{M \rightarrow M_0} f(M) = a$  iff for any point sequence  $\{M_n\} \subseteq D \setminus \{M_0\}$ , such that  $M_n \rightarrow M_0$  ( $n \rightarrow +\infty$ ), we have  $f(M_n) \rightarrow a$  ( $n \rightarrow +\infty$ )

$\lim_{M \rightarrow M_0} f(M)$  exists  $\iff \lim_{\delta \rightarrow 0} \sup_{B^-(M_0, \delta)} |f(M) - f(M')| = 0 \iff$  In particular,  $\lim_{\delta \rightarrow 0} \sup_{\partial B(M_0, \delta)} |f(M) - f(M')| = 0$

**Eg.**  $f(x, y) = \begin{cases} \frac{xy}{x^2+y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$



1. Along the ray  $y = kx$ ,  $f(x, kx) = \frac{k}{1+k^2} \rightarrow \frac{k}{1+k^2}$  ( $x \rightarrow 0$ )

$\implies$  The limits are different along different rays

$\implies \lim_{(x,y) \rightarrow (0,0)} f(x, y)$  does not exist

2.

$$\begin{aligned} \begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases} &\implies f(\delta \cos \theta, \delta \sin \theta) = \cos \theta \sin \theta \\ &\implies \lim_{\delta \rightarrow 0} \sup_{M, M' \in \partial B(0, \delta)} |f(M) - f(M')| = 1 \\ &\implies \lim_{(x,y) \rightarrow 0} f(x, y) \text{ does not exist} \end{aligned}$$

### 9.1.4 Continuity

**Def.** Let  $f : D (\subseteq \mathbb{R}^2) \rightarrow \mathbb{R}$  and  $M_0 \in D$ , we say  $f$  is continuous at  $M_0$  if  $\lim_{M \rightarrow M_0} f(M) = f(M_0)$ , i.e.  $\forall \varepsilon > 0, \exists \delta = \delta(\varepsilon) > 0$  s.t.  $\forall M \in D \cap B(M_0, \delta), |f(M) - f(M_0)| < \varepsilon$

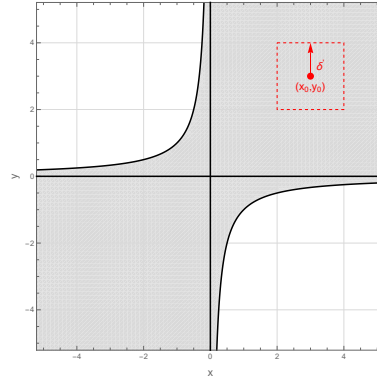
**Eg.**  $D = \{(x, y) \in \mathbb{R}^2 \mid xy > -1\}$ ,  $f(x, y) = \begin{cases} \log(1 + xy)^{\frac{1}{xy}} & xy \neq 0 \\ 1 & xy = 0 \end{cases}$

$$\forall \varepsilon > 0, \exists \delta = \delta(\varepsilon) > 0, \forall |u - x_0 y_0| < \delta$$

$$\text{Need } |xy - x_0 y_0| < \delta \implies |f(x, y) - f(x_0, y_0)| < \varepsilon$$

$$|xy - x_0 y_0| = |x_0| |y - y_0| - |y| |x - x_0|$$

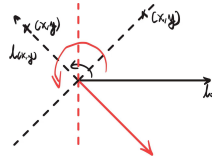
$$\begin{aligned} (|x_0| + 2|y_0| + 1)(|x - x_0| + |y - y_0|) &\geq |x_0| |y - y_0| - |y| |x - x_0| \\ &\geq |x_0 y - x_0 y_0| + |xy - x_0 y| \\ &\geq |f(x, y) - f(x_0, y_0)| \end{aligned}$$



$$\text{Take } \delta' = \min \left\{ \frac{\delta}{2(|x_0| + 2|y_0| + 1)} |y_0| \right\}.$$

$$\text{Then } \forall |x - x_0| < \delta', |y - y_0| < \delta', |xy - x_0 y_0| < \delta', |f(x, y) - f(x_0, y_0)| < \varepsilon$$

**Eg** (Angle Function).



$$\begin{cases} r = \sqrt{x^2 + y^2} \\ \theta = \text{the angle between the ray } l_0 \text{ and } l_{(x,y)} \text{ counted anti-clockwise} \end{cases}$$

$$(x, y) \in \mathbb{R}^2 \setminus \{(0, 0)\}$$

$$= \begin{cases} \arctan \frac{y}{x} & x > 0, y > 0 \\ \frac{\pi}{2} & x = 0, y > 0 \\ \pi + \arctan \frac{y}{x} & x < 0, y > 0 \\ \frac{3\pi}{2} & x = 0, y < 0 \\ 2\pi + \arctan \frac{y}{x} & x > 0, y < 0 \end{cases}$$

$$\begin{array}{c} \tan : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \xrightarrow{1:1} \mathbb{R} \\ \xrightarrow{\text{inverse}} \arctan : \mathbb{R} \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \end{array}$$

**Conclusion:**

1.  $\theta$  is continuous on  $\mathbb{R}^2 \setminus [0, +\infty)$  but has jumping discontinuity on  $(0, +\infty)$
2.  $r$  is continuous on  $\mathbb{R}^2$

**Def** (Uniform Continuity). Let  $f$  be defined on  $D \subseteq \mathbb{R}^2$ ,  $f$  is said to be uniformly continuous if  $\forall \varepsilon > 0, \exists \delta = \delta(\varepsilon) > 0$ , s.t.  $\forall M', M'' \in D, d(M', M'') < \delta, |f(M') - f(M'')| < \varepsilon$

**Thm.** 1. The continuity of function is preserved under  $+, -, \times$  and  $\div$  (provided the denominator function is nonzero at the point of interest)

2. The composition of continuous vector-valued functions is continuous.

3. (Intermediate Value Property) Let  $f$  be defined on a connected set  $D \subseteq \mathbb{R}^2$ , and  $f$  is continuous. Then for any  $M_1, M_2 \in D$ , all values between  $f(M_1)$  and  $f(M_2)$  can be achieved. In particular,  $f(D)$  is an interval. (More generally, the image of a connected set under a continuous vector-valued function is connected.)

**Pf.** Argue by contradiction.

Suppose  $f(D)$  is not connected. Then  $f(D) = A \sqcup B$ ,  $A, B \neq \emptyset$ ,  $A \cap \overline{B} = \emptyset$ ,  $B \cap \overline{A} = \emptyset$ .

$$\implies D = f^{-1}(A) \sqcup f^{-1}(B), f^{-1}(A), f^{-1}(B) \neq \emptyset$$

Suppose  $\{M_n\} \subseteq f^{-1}(A)$ ,  $M_n \rightarrow M_0 \in f^{-1}(B)$ , then  $f(\overset{A}{\downarrow} M_n) \rightarrow f(\overset{B}{\downarrow} M_0) \implies B \cap \overline{A} \neq \emptyset$ , contradiction.

$$\implies \overline{f^{-1}(A)} \cap f^{-1}(B) = \emptyset$$

Similarly,  $\implies \overline{f^{-1}(B)} \cap f^{-1}(A) = \emptyset$

$\implies$  This is a contradiction with the connectivity of  $D$ .

4.  $\overset{\text{supremum}}{\uparrow}$  The maximum and  $\overset{\text{infimum}}{\uparrow}$  minimum of a continuous function on a bounded closed set can be achieved.

**Pf.** Step 1)  $\sup_D f < +\infty, \inf_D f > -\infty$

If  $\sup_D f = +\infty$ , then  $\exists \{M_n\} \subseteq \overset{\text{closed, bounded}}{\downarrow} D$ , s.t.  $f(M_n) \rightarrow +\infty$

By Bolzano-Weierstrass Thm,  $\exists$  subsequence  $\{M_{n_k}\}$  s.t.  $M_{n_k} \rightarrow M_0 \in D (k \rightarrow +\infty)$ ,  $f(M_{n_k}) \rightarrow f(M_0) = +\infty (k \rightarrow +\infty)$  by continuity, contradiction.

Step 2)  $\sup_D f = f(M_0)$  for some  $M_0 \in D$

Suppose  $\sup_D f \neq f(M)$  for any  $M \in D$

$$\implies \exists \{M_n\} \subseteq D \text{ s.t. } f(M_0) \rightarrow \sup_D f$$

$$\stackrel{B.W.}{\implies} \exists \text{ subsequence } \{M_{n_k}\} \subseteq D, M_{n_k} \rightarrow M_0 \in D \implies f(M_{n_k}) \rightarrow f(M_0) = \sup_D f$$

5. (Cantor Thm) *Continuous function on a bounded closed set is uniformly continuous.*

## 9.2 多变量函数的微分

### 9.2.1 Differentiation

$$\int_b^a df = \int_{\partial[a,b]} f$$

$$\begin{aligned}\frac{\partial f}{\partial x}(x_0, y_0) &= \lim_{x \rightarrow x_0} \frac{f(x, y_0) - f(x_0, y_0)}{x - x_0} \\ \frac{\partial f}{\partial y}(x_0, y_0) &= \lim_{y \rightarrow y_0} \frac{f(x_0, y) - f(x_0, y_0)}{y - y_0}\end{aligned} \quad (\text{assume domain is open set})$$

are called the first order partial derivatives of  $f$  at  $(x_0, y_0)$ .

If  $\frac{\partial f}{\partial x}(x, y)$  exists, for any  $(x, y) \in D$ , then the function  $\frac{\partial f}{\partial x}$  is called the first order partial derivative (function), also denoted by  $f'_x, f'_y$

Similarly, we can define

$$\begin{aligned}\frac{\partial}{\partial x} \left( \frac{\partial f}{\partial x} \right) &= \frac{\partial^2 f}{\partial x^2} = f''_{xx} & \frac{\partial}{\partial y} \left( \frac{\partial f}{\partial y} \right) &= \frac{\partial^2 f}{\partial y^2} = f''_{yy} \\ \frac{\partial}{\partial y} \left( \frac{\partial f}{\partial x} \right) &= \frac{\partial^2 f}{\partial y \partial x} = f''_{xy} \quad \text{not necessarily} \neq \quad \frac{\partial}{\partial x} \left( \frac{\partial f}{\partial y} \right) = \frac{\partial^2 f}{\partial x \partial y} = f''_{yx}\end{aligned}$$

mixed 2nd-order partial derivatives

### Geometric Meaning of Second-order Partial Derivatives

$$\begin{aligned}\left. \frac{\partial^2 f}{\partial x^2} \right|_{(x_0, y_0)} &= \left. \frac{d}{dx} \right|_{x=x_0} \frac{\partial f}{\partial x}(x, y_0) = \left. \frac{d}{dx} \right|_{x=x_0} \left. \frac{d}{dt} \right|_{t=x} f(t, y_0) \\ &= \lim_{\Delta x \rightarrow 0} \frac{\frac{\partial f}{\partial x}(x_0 + \Delta x, y_0) - \frac{\partial f}{\partial x}(x_0, y_0)}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{1}{\Delta x} \lim_{\Delta t \rightarrow 0} \frac{f(x_0 + \Delta x + \Delta t, y_0) - f(x_0 + \Delta x, y_0) - f(x_0 + \Delta t, y_0) + f(x_0, y_0)}{\Delta t} \\ &= \lim_{\Delta x \rightarrow 0} \lim_{\Delta t \rightarrow 0} \frac{f(x_0 + \Delta x + \Delta t, y_0) - f(x_0 + \Delta x, y_0) - f(x_0 + \Delta t, y_0) + f(x_0, y_0)}{\Delta x \Delta t}\end{aligned}$$

↓  
The second order ordinary derivative of the function  $f(x, y_0)$  at  $x_0$

$$\begin{aligned}\left. \frac{\partial^2 f}{\partial y \partial x} \right|_{(x_0, y_0)} &= \lim_{\Delta y \rightarrow 0} \frac{\frac{\partial f}{\partial x}(x_0, y_0 + \Delta y) - \frac{\partial f}{\partial x}(x_0, y_0)}{\Delta y} \\ &= \lim_{\Delta y \rightarrow 0} \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0 + \Delta y) - f(x_0 + \Delta x, y_0) + f(x_0, y_0)}{\Delta x \Delta y}\end{aligned}$$

↓  
 $\frac{\Pi(\Delta x, \Delta y)}{\Delta x \Delta y}$

$$\left. \frac{\partial^2 f}{\partial x \partial y} \right|_{(x_0, y_0)} = \lim_{\Delta x \rightarrow 0} \lim_{\Delta y \rightarrow 0} \frac{\Pi(\Delta x, \Delta y)}{\Delta x \Delta y}$$

**Thm.** If  $\frac{\partial^2 f}{\partial x \partial y}, \frac{\partial^2 f}{\partial y \partial x}$  exist in an open neighborhood of  $(x_0, y_0)$  and are both continuous at  $(x_0, y_0)$ , then  $\left. \frac{\partial^2 f}{\partial y \partial x} \right|_{(x_0, y_0)} = \left. \frac{\partial^2 f}{\partial x \partial y} \right|_{(x_0, y_0)}$

**Pf.**

$$\begin{aligned}
& f(x_0 + h, y_0 + k) - f(x_0, y_0 + k) - f(x_0 + h, y_0) + f(x_0, y_0) \\
&= \phi(h) - \phi(0) \text{ where } \phi(h) = f(x_0 + h, y_0 + k) - f(x_0 + h, y_0) \\
& \quad \phi(0) = f(x_0, y_0 + k) - f(x_0, y_0) \\
& \stackrel{\exists \theta \in (0,1)}{\underset{L.M.V.T.}{=}} \phi'(\theta h)h = \left( \frac{\partial f}{\partial x}(x_0 + \theta h, y_0 + k) - \frac{\partial f}{\partial x}(x_0 + \theta h, y_0) \right) h \\
& \stackrel{\exists \eta \in (0,1)}{\underset{L.M.V.T.}{=}} \frac{\partial}{\partial y} \left( \frac{\partial f}{\partial x} \right) \Big|_{(x_0 + \theta h, y_0 + \eta k)} kh
\end{aligned}$$

$$\begin{aligned}
& \text{Similarly, } \Pi(h, k) = \frac{\partial^2 f}{\partial x \partial y}(x_0 + \theta h, y_0 + \eta k)hk = \frac{\partial^2 f}{\partial y \partial x}(x_0 + \theta h, y_0 + \eta k)hk \\
& \text{Take limit } h, k \rightarrow 0 \implies \frac{\partial^2 f}{\partial x \partial y} \Big|_{(x_0, y_0)} = \frac{\partial^2 f}{\partial y \partial x} \Big|_{(x_0, y_0)}
\end{aligned}$$

### 9.2.2 多变量函数的可微性

### 9.2.3 方向导数与梯度

### 9.2.4 复合函数的微分和一阶微分形式不变性

### 9.2.5 向量值函数的微商和微分

## 9.3 隐函数定理和逆映射定理

### 9.3.1 隐函数的存在性和微商

### 9.3.2 从微分的角度看隐函数定理

### 9.3.3 逆映射的微商

## 9.4 空间曲线与曲面

### 9.4.1 参数曲线

### 9.4.2 参数曲面

### 9.4.3 隐式曲线和隐式曲面

## 9.5 多变量函数的 Taylor 公式与极值

### 9.5.1 二元函数的微分中值定理

### 9.5.2 二元函数的 Taylor 公式

笔记

### 9.5.3 二元函数的极值

笔记

### 9.5.4 条件极值

笔记

## 9.6 向量场的微商

### 9.6.1 向量场

### 9.6.2 梯度、散度与旋度

### 9.6.3 Hamilton 算子在柱面坐标系和球面坐标系中的表示

## 9.7 微分形式

### 9.7.1 微分形式的空间

### 9.7.2 微分形式的外积

### 9.7.3 微分形式的外微分

### 9.7.4 微分形式在高维空间的推广

## 10 多变量函数的重积分

### 10.1 二重积分

#### 10.1.1 平面区域的面积

#### 10.1.2 二重积分的基本概念与性质

#### 10.1.3 二重积分的计算

### 10.2 二重积分的换元

#### 10.2.1 坐标曲线和面积元素

#### 10.2.2 二重积分的换元

### 10.3 三重积分

#### 10.3.1 三重积分的累次积分

#### 10.3.2 三重积分的换元

#### 10.3.3 来自物理学中的几个例子

### 10.4 $n$ 重积分

## 11 曲线积分和曲面积分

### 11.1 数量场在曲线上的积分

#### 11.1.1 基本概念

#### 11.1.2 数量场在曲线上的积分的计算

### 11.2 数量场在曲面上的积分

#### 11.2.1 曲面的面积

#### 11.2.2 数量场在曲面上的积分的计算

### 11.3 向量场在曲线上的积分

#### 11.3.1 曲线的定向

#### 11.3.2 向量场在曲线上的积分的定义和计算

#### 11.3.3 Green 定理

### 11.4 向量场在曲面上的积分

#### 11.4.1 双侧曲面及其定向

#### 11.4.2 向量场在曲面上的积分的定义和计算

### 11.5 Gauss 定理和 Stokes 定理

#### 11.5.1 Gauss 定理

#### 11.5.2 Stokes 定理

### 11.6 其他形式的曲线、曲面积分

### 11.7 保守场

#### 11.7.1 保守场与势函数

#### 11.7.2 无源场与向量势

### 11.8 微分形式的积分

#### 11.8.1 微分形式的积分

#### 11.8.2 全微分方程

## 12 Fourier 分析

### 12.1 函数的 Fourier 级数

#### 12.1.1 周期函数与三角函数的正交性

#### 12.1.2 周期函数的 Fourier 级数

#### 12.1.3 有限区间上函数的 Fourier 级数

#### 12.1.4 Fourier 级数的复数形式

### 12.2 平方平均收敛

#### 12.2.1 基本概念

#### 12.2.2 Bessel (贝塞尔) 不等式

#### 12.2.3 平方平均收敛

#### 12.2.4 广义 Fourier 级数

### 12.3 收敛性定理的证明

#### 12.3.1 Dirichlet 定理的证明

#### 12.3.2 平方平均收敛性定理的证明

### 12.4 Fourier 变换

#### 12.4.1 Fourier 积分

#### 12.4.2 Fourier 变换

## 13 反常积分和含参变量的积分

### 13.1 反常积分

#### 13.1.1 无穷区间上积分的收敛性

#### 13.1.2 无穷区间上积分收敛性的一般判别法

#### 13.1.3 无界函数积分的收敛判别法

### 13.2 反常多重积分

### 13.3 含参变量的积分

#### 13.3.1 含参变量的积分及其性质

#### 13.3.2 积分限依赖于参变量的积分及其性质

### 13.4 含参变量的反常积分

#### 13.4.1 含参变量的反常积分的一致收敛性

#### 13.4.2 含参变量反常积分的性质

#### 13.4.3 几个重要的积分

### 13.5 Euler 积分

#### 13.5.1 $\Gamma$ 函数的性质

#### 13.5.2 $B$ 函数的性质

**Def** (定义标题). 定义内容...

**Thm** (定理标题). 定理内容...

证明. 证明过程...

□

**Eg** (例题标题). 例题内容...

**Rmk.** 备注说明...

## 13.6 重要公式

...

(1)

## 13.7 课后总结

## 14 第 $X+1$ 章：[章节标题]